chromatids) are drawn to opposite poles of the cell, a process termed **disjunction**. Thus the chromosome number is halved, and each cellular product of meiosis I has the haploid chromosome number. The 23 pairs of homologous chromosomes assort independently of one another, and as a result, the original paternal and maternal chromosome sets are sorted into random combinations. The possible number of combinations of the 23 chromosome pairs that can be present in the gametes is 2^{23} (more than 8 million). Owing to the process of crossing over, however, the variation in the genetic material that is transmitted from parent to child is actually much greater than this. As a result, each chromatid typically contains segments derived from each member of the original parental chromosome pair, as illustrated schematically in Figure 2-14. For example, at this stage, a typical large human chromosome would be composed of three to five segments, alternately paternal and maternal in origin, as inferred from DNA sequence variants that distinguish the respective parental genomes (Fig. 2-15).

After telophase of meiosis I, the two haploid daughter cells enter meiotic interphase. In contrast to mitosis, this interphase is brief, and meiosis II begins. The notable point that distinguishes meiotic and mitotic interphase is that there is no S phase (i.e., no DNA

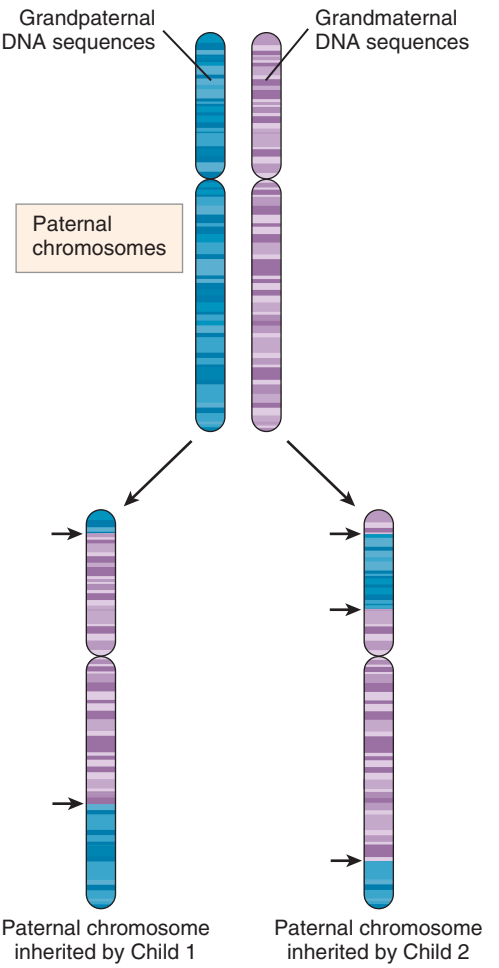


Figure 2-15 The effect of homologous recombination in meiosis. In this example, representing the inheritance of sequences on a typical large chromosome, an individual has distinctive homologues, one containing sequences inherited from his father (*blue*) and one containing homologous sequences from his mother (*purple*). After meiosis in spermatogenesis, he transmits a single complete copy of that chromosome to his two offspring. However, as a result of crossing over (*arrows*), the copy he transmits to each child consists of alternating segments of the two grandparental sequences. Child 1 inherits a copy after two crossovers, whereas child 2 inherits a copy with three crossovers.

GENETIC CONSEQUENCES AND MEDICAL RELEVANCE OF HOMOLOGOUS RECOMBINATION

The take-home lesson of this portion of the chapter is a simple one: **the genetic content of each gamete is unique,** because of random assortment of the parental chromosomes to shuffle the combination of sequence variants *between* chromosomes and because of homologous recombination to shuffle the combination of sequence variants *within* each and every chromosome. This has significant consequences for patterns of genomic variation among and between different populations around the globe and for diagnosis and counseling of many common conditions with complex patterns of inheritance (see Chapters 8 and 10).

The **amounts and patterns of meiotic recombination** are determined by sequence variants in specific genes and at specific "hot spots" and differ between individuals, between the sexes, between families, and between populations (see Chapter 10).

Because recombination involves the physical intertwining of the two homologues until the appropriate point during meiosis I, it is also critical for ensuring proper chromosome segregation during meiosis. Failure to recombine properly can lead to **chromosome missegregation** (**nondisjunction**) in meiosis I and is a frequent cause of pregnancy loss and of chromosome abnormalities like Down syndrome (see Chapters 5 and 6).

Major ongoing efforts to **identify genes and their variants responsible for various medical conditions** rely on tracking the inheritance of millions of sequence differences within families or the sharing of variants within groups of even unrelated individuals affected with a particular condition. The utility of this approach, which has uncovered thousands of gene-disease associations to date, depends on patterns of homologous recombination in meiosis (see Chapter 10).

Although homologous recombination is normally precise, areas of repetitive DNA in the genome and genes of variable copy number in the population are prone to occasional **unequal crossing over** during meiosis, leading to variations in clinically relevant traits such as drug response, to common disorders such as the thalassemias or autism, or to abnormalities of sexual differentiation (see Chapters 6, 8, and 11).

Although homologous recombination is a normal and essential part of meiosis, it also occurs, albeit more rarely, in somatic cells. Anomalies in **somatic recombination** are one of the causes of **genome instability in cancer** (see Chapter 15).